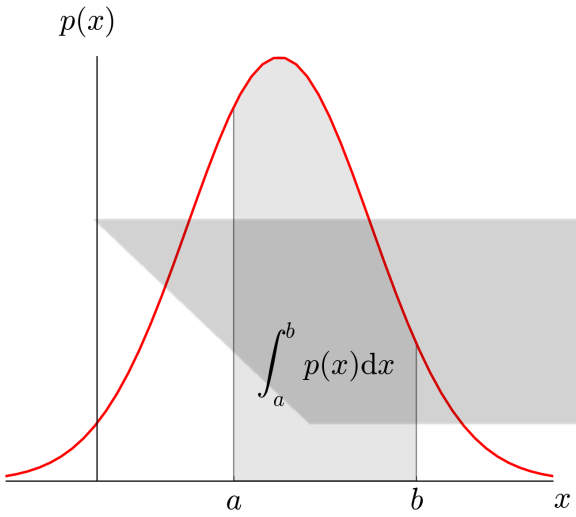
1. Continuous Random Variables

1.1. Probability Density Functions (pdf)

- Function whose area under its graph represents probability used for continuous random variables
- Represented by $f(x)$



Conditions:

- Total area always $=1$

$$\int_c^d f(x)\,dx = 1$$

- Cannot have negative probabilities $\therefore$ graph cannot dip below x-axis; $f(x) \geq 0$
- Probability that X lies between a and b is the area from a to b

$$P(a < X < b) = \int_a^b f(x)\,\mathrm{dx}$$

- Outside given interval $f(x) = 0$; show on a sketch
- $P(X = b)$ always equals 0 as there is no area
- Notes:
  - $P(X < b) = P(X \leq b)$ as no extra area added
  - The mode of a pdf is its maximum (stationary point)

{IS Ex 9a} Question 6:

Given that:

$$f(x) = \begin{cases} kx(6-x) & 2 < x < 5 \\ 0 & \text{otherwise} \end{cases}$$

1. Find the value of k
2. Find the mode, m
3. Find $P(X < m)$

Solution:

Part (i):

Total area must equal 1 hence

$$\int_2^5 \mathrm{kx}(6-x) = \left[3kx^2 - \frac{kx^3}{3}\right]_2^5 = 1$$

$$= 75k - \frac{125}{3}k - 12k + \frac{8}{3}k = 24k = 1$$

$$\therefore k = \frac{1}{24}$$

Part (ii):

Mode is the value which has the greatest probability hence we are looking for the max point on the pdf

$$\frac{d}{\mathrm{dx}}[\mathrm{kx}(6-x)] = 6k - 2kx$$

Finding max point hence stationary point

$$6k - 2kx = 0$$

$$x = \frac{6\left(\frac{1}{24}\right)}{2\left(\frac{1}{24}\right)} = 3$$

$\therefore$ mode $=3$

Part (iii):

$P(X < m)$ *can be interpreted as* $P(-\infty < X < m)$

$$\int_{-\infty}^m \mathrm{kx}(6-x) = \int_2^3 \mathrm{kx}(6-x) = \left[3kx^2 - \frac{kx^3}{3}\right]_2^3$$

$$= \frac{1}{24}\left(3(3^2) - \frac{3^3}{3} - 3(2^2) + \frac{2^3}{3}\right) = \frac{13}{36}$$

1.2. Mean & Variance

- To calculate mean/expectation

$$E(X) = \int_{-\infty}^{\infty} \mathrm{xf}(x)\,\mathrm{dx}$$

- To calculate variance:
  - First calculate $E(X)$ then $E(X^2)$ by

$$E\left(X^2\right) = \int_{-\infty}^{\infty} x^2 f(x)\,\mathrm{dx}$$

- Substitute information and calculate using

$$\mathrm{Var}(X) = E\left(X^2\right) - E(X)^2$$

1.3. The Median

The Cumulative Distribution Function (cdf)

- Gives the probability that the value is less than b

$P(X < b)$ or $P(X \leq b)$

- Represented by $F(b)$
- It is the integral of $f(x)$

$$F(b) = \int_{-\infty}^b f(x)\,\mathrm{dx}$$

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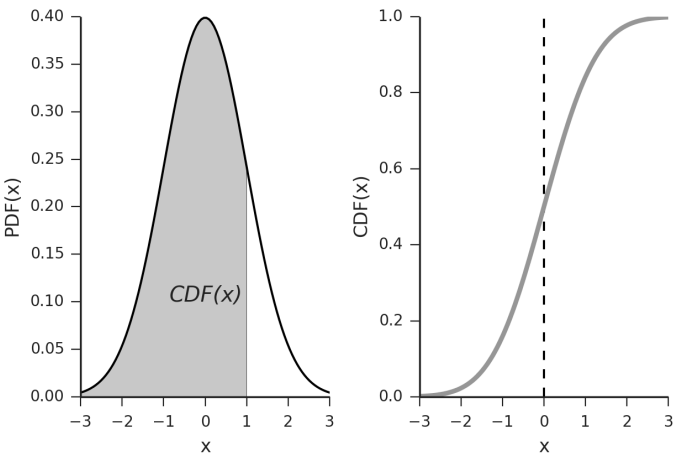
- Median: the value of b for which $F(b) = 0.5$



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